ASSESSING DEHYDRATION FLUID THERAPY

THE GOAL OF FLUID THERAPY

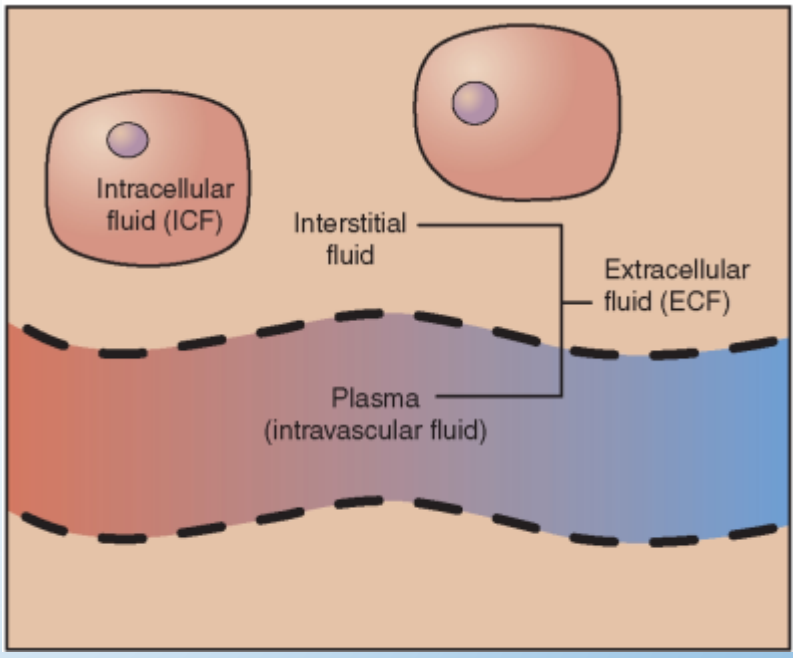
- REHYDRATION
- REPLACEMENT
- RESUSCITATION

WHAT ARE THE INDICATIONS FOR FLUID THERAPY?

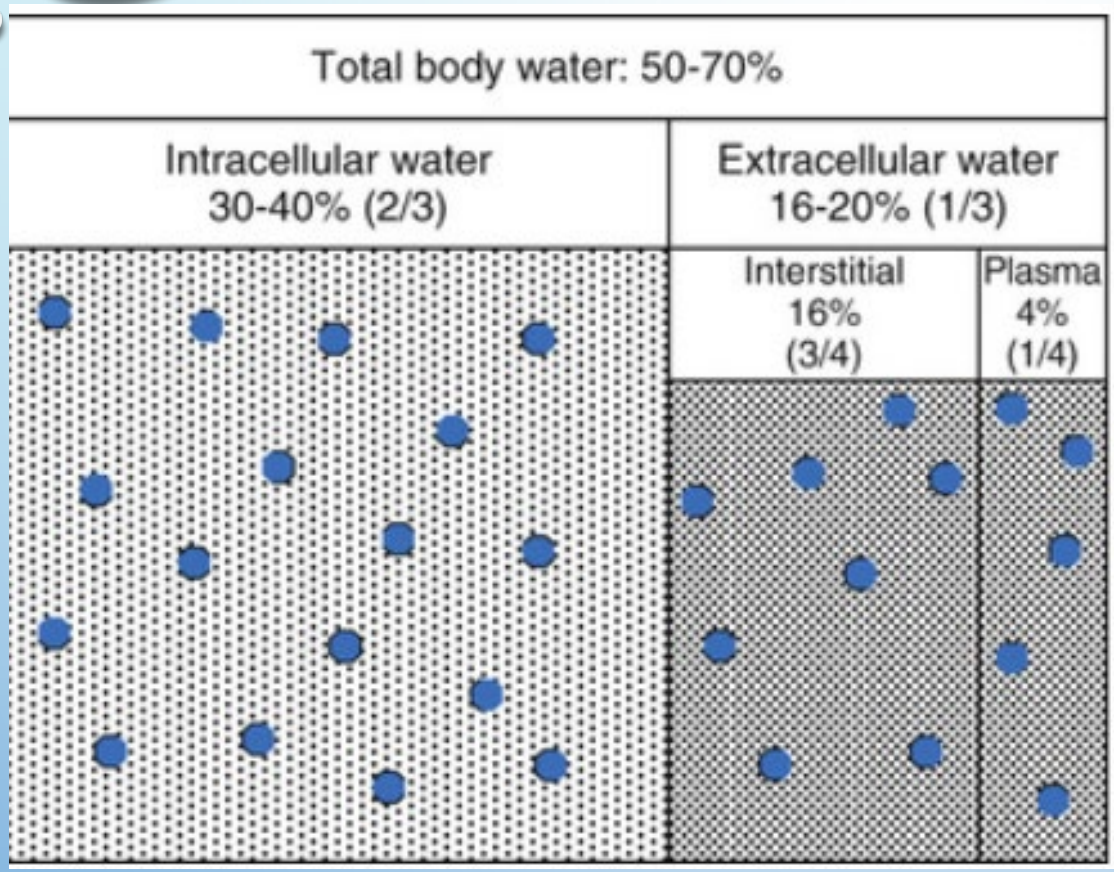
- MAINTAINING HYDRATION
- REPLACING FLUID DEFICIT (DEHYDRATION)
- REPLACING ONGOING LOSSES
- TREATING DECREASED ONCOTIC PRESSURE
- TREATING HYPOVOLEMIA
- TREATING SHOCK STATES
- IMPROVING OR INCREASING URINE PRODUCTION
- CORRECTING ACID/BASE BALANCE OR ELECTROLYTE DISORDERS
- MAINTAINING INTRAVENOUS ACCESS AND DELIVERING OTHER MEDICATIONS

BODY FLUIDS

- **FLUID COMPARTMENTS**
  - **INTRACELLULAR FLUID – ICF**
    - **FLUID WITHIN ALL OF THE CELLS OF THE BODY**
  - **EXTRACELLULAR FLUID – ECF**
    - **INTERSTITIAL FLUID** – FLUID BETWEEN THE CELLS OF TISSUE
    - **INTRAVASCULAR FLUID** = PLASMA – THE LIQUID COMPONENT OF BLOOD AND LYMPH IN WHICH ERYTHROCYTES , LEUKOCYTES AND PLATELETS ARE SUSPENDED



- The adult animal's body is 55-65% Water by weight
- Young animals body is up to 85% water by weight



What are the barriers between compartments?

ICF/ECF barrier = the cell membrane

Interstitial/Intravascular barrier = vascular endothelium and glycocalyx (protein/lipid layer on endothelial surface)

WATER BALANCE

NATURAL WATER LOSSES

SENSIBLE	INSENSIBLE
(MEASURABLE)	(NOT MEASURABLE)
URINATION	DEFECATION
	RESPIRATION (EXPIRATION)
	SWEAT

Contemporary (ongoing) water losses, such as:

Diarrhea

Vomiting

Hemorrhage

Sialorrhea

Hyperhidrosis

* Circulating blood volume
  * Dogs = 80-90 ml/kg
  * Cats = 40-60 ml/kg
  * Horses = 80 ml/kg

Also, pyrexia will cause the body to burn its water faster.

WATER BALANCE

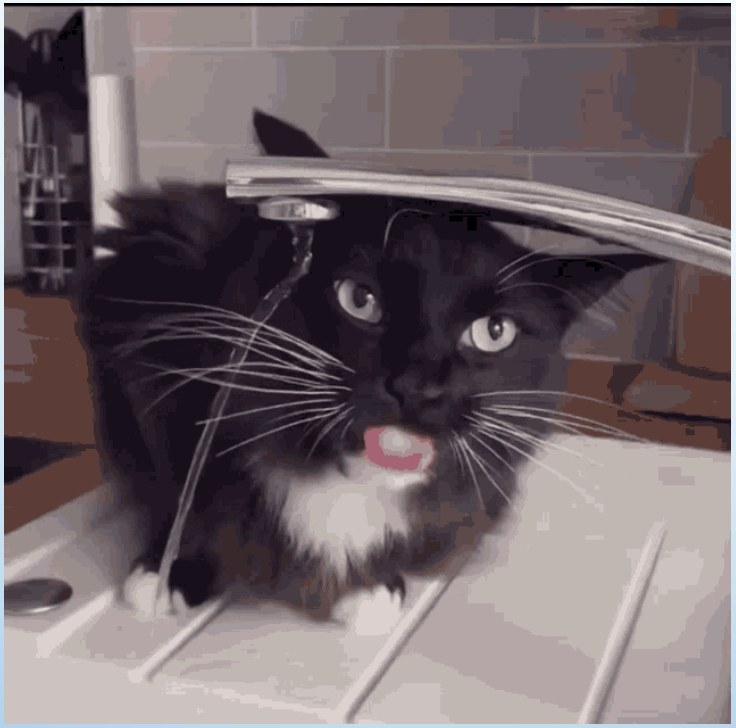
DEHYDRATION: WATER LOSS > WATER GAIN

- DECREASED BLOOD VOLUME:
  - DECREASED BLOOD PRESSURE
  - REDUCED TISSUE PERFUSION.
- ELECTROLYTE LOSS
  - USUALLY RESULTS IN METABOLIC ACIDOSIS

10% LOSS OF BODY WATER = SERIOUS METABOLIC DISTURBANCES

30% LOSS OF BODY WATER = DEATH

ASSESSING DEHYDRATION



Nursing Assessments

OBTAINING A HISTORY

- NO FOOD OR WATER INTAKE FOR THE PAST 24 HOURS
- PRESENCE OF VOMITING, DIARRHEA OR INCREASED VOLUME OR FREQUENCY OF URINATION
- WHAT QUESTIONS WOULD YOU ASK?

INTRAVASCULAR LOSS

- Hypotension
- Tachycardia
- Obtundation
- Hypothermia
- Cool extremities
- Prolonged CRT
- Bounding/ thread pulses
- Weight loss
- BUT WHY??

INTERSTITIAL LOSS

SKIN TURGOR

ALWAYS USE THE SAME SPOT.

CRT

Weight

Heart Rate
Pulses

Mucous Membranes

Sunken eyes

Elevated PCV/ TS

Full exam required
Animals can be dehydrated and have normal intravascular volume
They can also have normal hydration and be hypovolemic
Geriatric patients

EVALUATING DEHYDRATION

ESTIMATED % DEHYDRATION	PHYSICAL EXAMINATION FINDING
<5%	• Not detectable
5–6%	• Some change in skin turgor
6–8%	• Mild decreased skin turgor • Dry mucous membranes*
8–10%	• Obvious decreased skin turgor • Retracted globes within orbits
10–12%	• Persistent skin tent due to complete loss of skin elasticity • Dull corneas** • Evidence of hypovolemia
>12%	• Hypovolemic shock • Death

2024 AAHA Fluid Guidelines

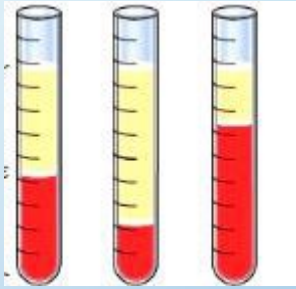
NURSING ASSESSMENTS

- Altered electrolyte/ chemistry status
- Altered erythrocyte status
- Altered gas exchange
- Altered mental status
- Altered urination
- Constipation
- Fluid volume deficit
- Impaired self care
- Impaired thermoregulation
- Reduced cardiac output
- Reduced perfusion

PACKED CELL VOLUME & TOTAL PROTEIN/ SOLID

IF WATER INTAKE IS REDUCED OR WATER EXCRETION INCREASED AND, THERE IS A NET LOSS OF BODY WATER

DEHYDRATION = LOSS OF BODY WATER CONCENTRATES THE CELLS AND PROTEINS SUSPENDED IN THE BLOOD



THEREFORE, PCV AND TP CAN BE USED TO ASSESS HYDRATION STATUS.

NORMAL	PCV	TP/TS
DOG	37 - 55 %	6-8 G/DL
CAT	24 - 55 %	5.7 – 8.9 G/DL

NOTE: THESE ARE THE IDEXX VALUES- OTHER SOURCES MAY VARY

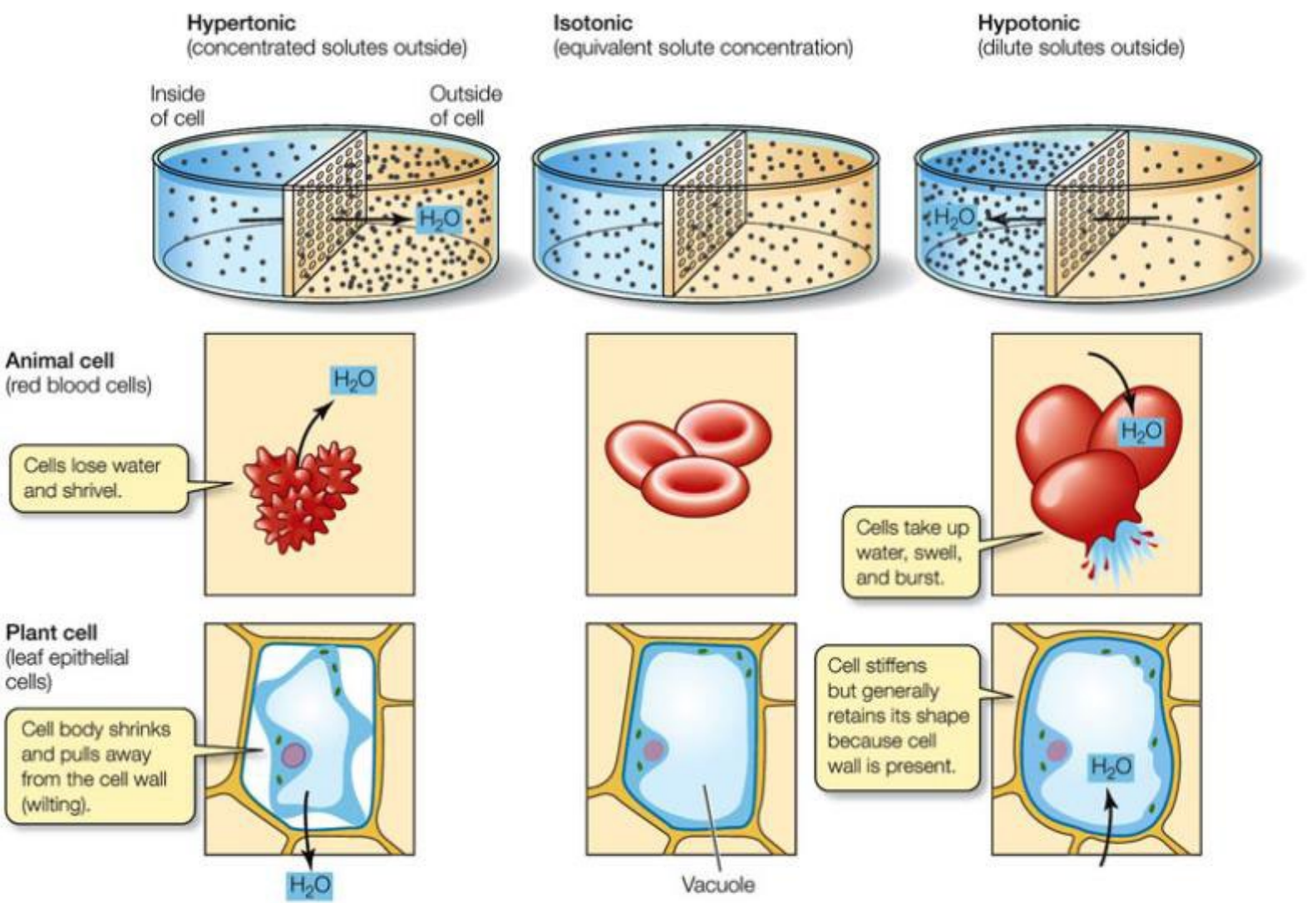
- ***MAINTAIN BLOOD VOLUME***
- REPLACEMENT OF WATER LOSSES DUE TO ILLNESS OR INJURY
- PREVENT DEHYDRATION
- CORRECT ACID/BASE (BLOOD PH) DISTURBANCES
- CORRECT OR PREVENT ELECTROLYTE IMBALANCES
  - NA^+, K^+, CL^+, ETC., FROM THE BODY.

THERAPEUTIC ADMINISTRATION OF FLUIDS

FLUID THERAPY

- KEYS TO FLUID THERAPY PLANNING
  - RIGHT TYPE
  - RIGHT AMOUNT
  - RIGHT SPEED
- CONSIDERATIONS
  - BODY FLUID COMPARTMENTS
  - FLUID TYPES
  - RATES AND ROUTES
  - EFFECTIVENESS MONITORING
  - POTENTIAL COMPLICATIONS

TONICITY

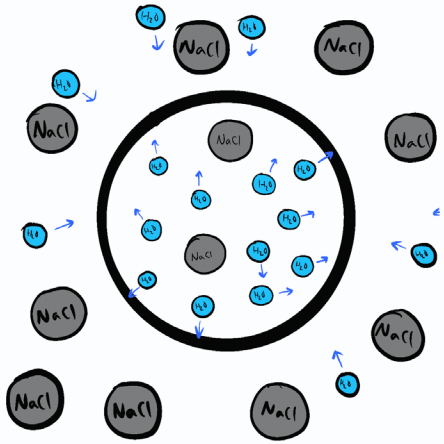


LIFE 8e, Figure 5.9

LIFE: THE SCIENCE OF BIOLOGY, Eighth Edition © 2007 Sinauer Associates, Inc. and W. H. Freeman & Co.

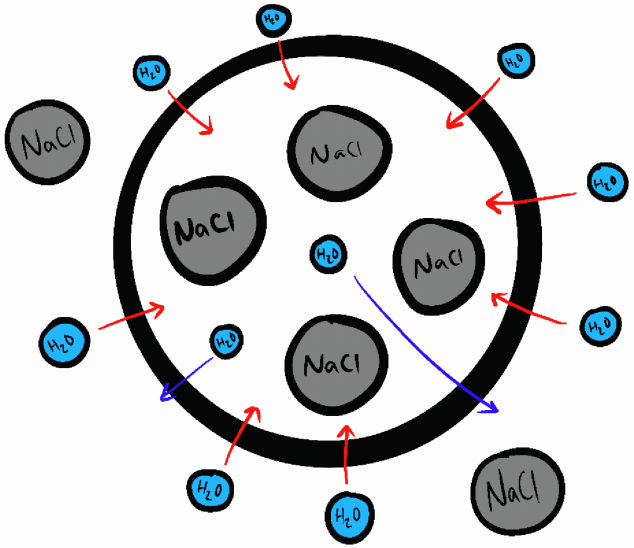
Hypertonic Fluid

Outside cell: Fluid has many molecules of solute in it.
- low concentration of water molecules
- Solute can't pass cell membrane to reach equilibrium
- Water can go out of cell to reach equilibrium
  - ↳ will rush out
  - ↳ some will come in but too slow
  - ↳ net movement of water will move outwards
    - cell will shrivel Plasmolysis



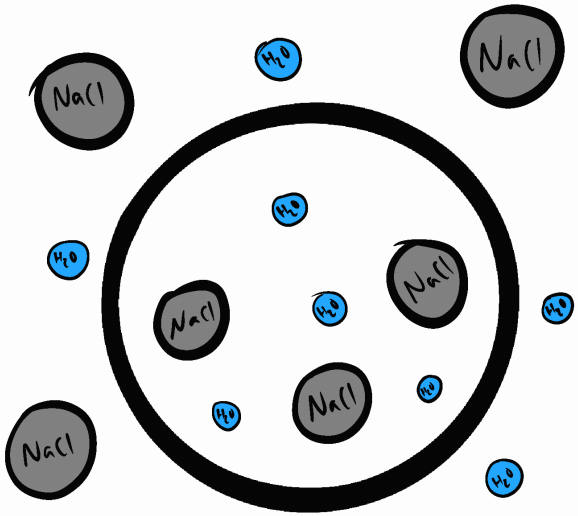
Hypotonic Solution

- Cell emerged in less concentrated solution
- outside cell: more H_2O than NaCl
- inside cell: more NaCl than H_2O
- Some H_2O leaves cell but more H_2O rush into the cell b/c not obstructed by solute particles
  - ~ cell's size increases due to too much H_2O
    - Cytolysis (bursts)



Isotonic Solution

- Concentrations inside & outside the cell are equal
- Net movement is 0 allowing H_2O to enter and leave the cell in equal amounts



WE NEED TO TAKE THE PATIENTS CONDITION INTO CONSIDERATION

- If a patient is dehydrated what would happen if we give them
  - Hypertonic solution
  - Hypotonic solution
  - Isotonic solution
- If a patient is overhydrated what would happen if we give them
  - Hypertonic solution
  - Hypotonic solution
  - Isotonic solution
- If a patient is appropriately hydrated what would happen if we give them
  - Hypertonic solution
  - Hypotonic solution
  - Isotonic solution

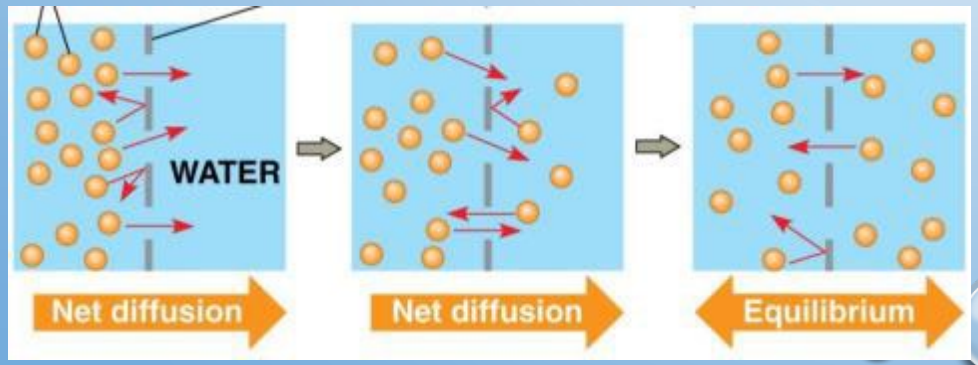
DIFFUSION ABILITY

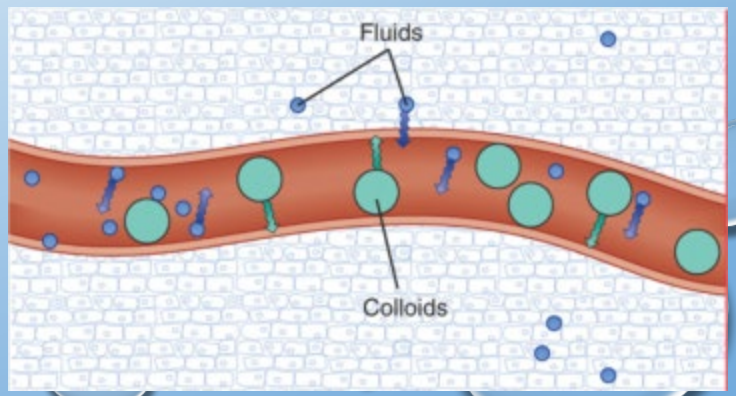
CRYSTALLOIDS

- SOLUTIONS CONTAINING SUBSTANCES THAT PASS EASILY THROUGH CELL MEMBRANES.
  - ELECTROLYTES

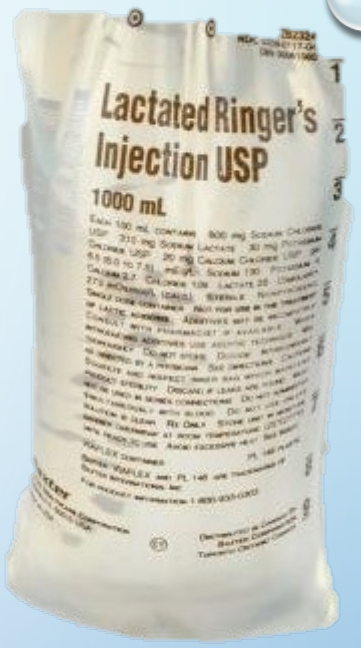
COLLOIDS

- SOLUTIONS THAT CONTAIN LARGE MOLECULES





COMMON CRYSTALLOIDS



- LRS: NA^+, CL^-, CA^{++}, K^+, LACTATE
  - MOST COMMONLY USED
  - MAINTENANCE OF NORMAL HYDRATION
  - CORRECTION OF DEHYDRATION
  - REPLACEMENT OF COMMONLY DEPLETED ELECTROLYTES (VOMITING, DIARRHEA, RENAL DISEASE)
  - TX OF SHOCK
  - CORRECTION OF METABOLIC ACIDOSIS
    - <u>HEALTHY LIVERS WILL CONVERT LACTATE INTO BICARBONATE</u>

COMMON CRYSTALLOIDS

- NORMOSOL-R, MULTISOL-R, PLASMA-LYTE: NA^+, CL^-, CA^{++}, K^+, MG
  - SIMILAR TO LRS WITH SIMILAR USES. PREFERRED WITH PATIENTS WITH LIVER DZ BECAUSE ACETATE AND GLUCONATE ARE CONVERTED TO PH BUFFERS IN THE MUSCLE RATHER THAN THE LIVER

COMMON CRYSTALLOIDS

SALINE: NA^+, CL^-

- SODIUM IS THE MAJOR SOLUTE IN EXTRACELLULAR SPACE
- SODIUM BASED SOLUTIONS RAPIDLY DISTRIBUTE, AND INCREASE THE VOLUME OF BOTH PLASMA AND INTERSTITIAL FLUID.

DEXTROSE: GLUCOSE

- USED TO TREAT HYPOGLYCEMIA, ESPECIALLY NEONATES. 5% IS ISOTONIC WHEN ADMINISTERED, BUT ONCE GLUCOSE IS METABOLIZED, THE REMAINING WATER IS HYPOTONIC. CAN DEPLETE ELECTROLYTES IN THE BODY, DUE TO EXCRETION OF WATER THROUGH THE KIDNEYS.

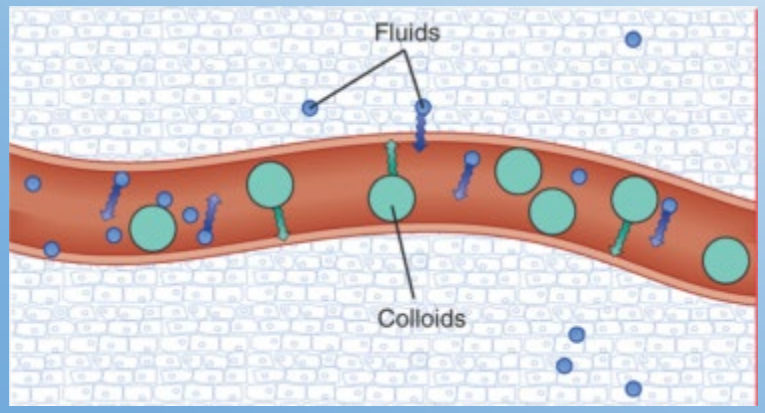
a. <u>5% DEXTROSE MAY BE “SPIKED” WITH ADDITIONAL ELECTROLYTES</u>

b. <u>ADMINISTER DEXTROSE/ SALINE OR DEXTROSE/LRS COMBINATIONS</u>

c. <u>INTERMITTENTLY ADMINISTERED HYPERTONIC DEXTROSE 10% OR 50% SLOW IV</u>

COLLOIDS

- **HYPERTONIC** SOLUTIONS THAT CONTAIN LARGE MOLECULES
  - CANNOT PASS THROUGH CELL MEMBRANE
  - REFERRED TO AS "VOLUME EXPANDERS"
  - COLLOIDS REMAIN IN PLASMA AND SERVE AS OSMOTICALLY ACTIVE PARTICLES
    - COLLOIDS WILL HOLD FLUID IN THE INTRAVASCULAR SPACE
    - DRAW FLUID FROM INTERSTITIAL SPACE INTO THE VASCULAR SPACE TO INCREASE PLASMA VOLUME
    - COLLOIDS ARE 5-10 TIMES MORE EFFECTIVE THAN CRYSTALLOIDS AT INCREASING PLASMA VOLUME.



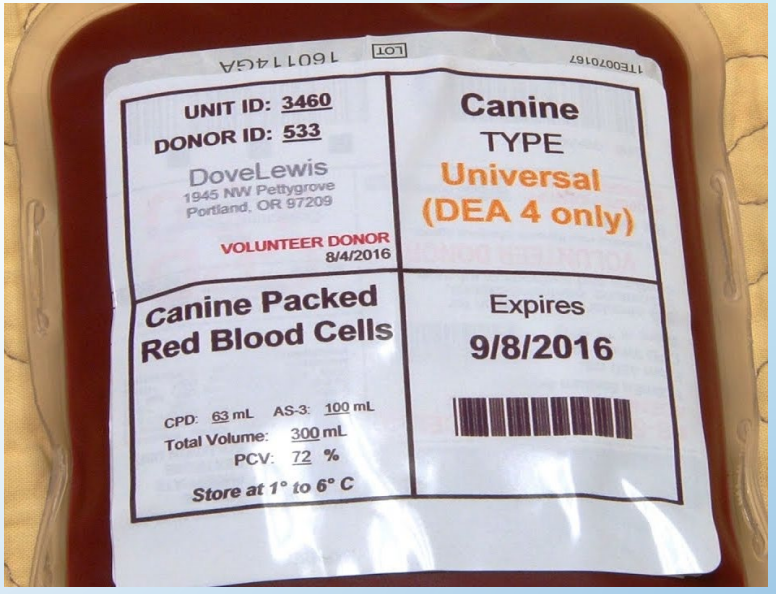
COLLOIDS

- ALBUMIN IS AN OSMOTICALLY ACTIVE PROTEIN COMMONLY FOUND IN BLOOD. COLLOIDS SERVE AS A REPLACEMENT FOR ALBUMIN TO TREAT *HYPOPROTEINEMIA* AND ASSOCIATED SYMPTOMS (ASCITES, EDEMA).

- COLLOIDS MAY BE USED ALONE OR IN COMBINATION WITH CRYSTALLOIDS
  - TREATS *HYPOVOLEMIA* RESULTING FROM BLOOD LOSS

COMMON COLLOIDS

- PLASMA
- WHOLE BLOOD
- PACKED RED BLOOD CELLS
- HYDROXYETHYL STARCHES DERIVED FROM CORN
  - WILL REMAIN IN PLASMA 12-48 HOURS
  - DOES NOT CONTAIN RBC AND CLOTTING FACTORS
- DEXTRANS
  - 4-8 HRS.



SELECTION OF FLUIDS

- ADMINISTERING COLLOIDS WITH CRYSTALLOIDS REDUCES THE NEED FOR CRYSTALLOIDS 40%-60%
  - FLUID OVERLOAD LESS LIKELY
  - RESUSCITATION TIMES ARE SHORTER
- MANY CONDITIONS CAN CAUSE INCREASED CAPILLARY PERMEABILITY
  - PARVO, SEVERE GI DISEASE, PANCREATITIS, SEPTIC SHOCK, MASSIVE TRAUMA, HEAT STROKE, AND OTHERS.
  - HETASTARCH OR STROMA-FREE HEMOGLOBIN ARE THE CHOICE FOR RESUSCITATION

ROUTES OF ADMINISTRATION

ORAL- PO

> NATURAL INTAKE PREFERRED FOR FLUID MAINTENANCE

> **CAN NOT** BE USED WITH PATIENTS WITH:

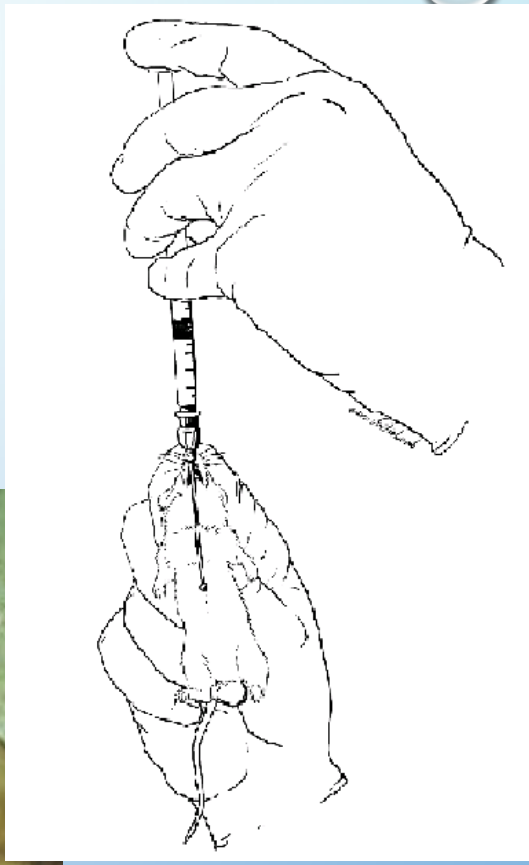
- VOMITING
- MALABSORBTION
- GI OBSTRUCTION
- OR RAPID IV FLUID VOLUME IS NEEDED (SHOCK)

> COMMERCIALLY AVAILABLE ELECTROLYTE SOLUTIONS ARE PREFERRED

ENTERAL ADMINISTRATION

Enteral

- Gavage tubes and needles are commonly used with rodents, reptiles, and birds.
- Nasogastric tubes can be used in rabbits
- Esophagostomy tubes are surgically placed and can be used long term.



SUBCUTANEOUS- SC OR SQ

- EASY TO ADMINISTER
- FLUIDS GRADUALLY ABSORBED
- COMMONLY USED WITH BIRDS AND SMALL MAMMALS, WHEN IV IS DIFFICULT
- FLUIDS SHOULD BE ISOTONIC.
  - HYPERTONIC SOLUTIONS WILL DEHYDRATE THE SURROUNDING TISSUES
  - POORLY ABSORBED (6-8HRS.)
  - DEXTROSE SHOULD BE AVOIDED SINCE IT WILL ALSO PROMOTE BACTERIAL GROWTH
- 50-150 ML PER SITE

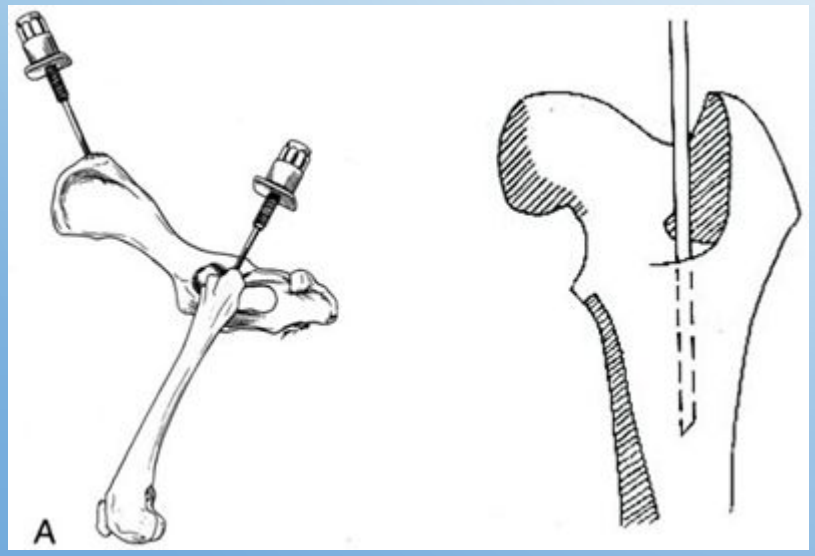
INTRAPERITONEAL- IP (MAMMALS) OR INTRACOELOMIC (REPTILES)

- ALTERNATIVE TO SQ WHEN LARGER VOLUMES ARE NECESSARY
- COMMONLY USED WITH REPTILES AND LAB ANIMALS
- NOT USED IN BIRDS DUE TO THE DISTRIBUTION OF AIR SACS

- **PERITONEAL DIALYSIS**
  - CATHETER IS PLACED THROUGH THE ABDOMINAL WALL (STRICT ASEPSIS), WARM DIALYSATES ARE INFUSED AND REMAIN 30-45MINUTES. TOXINS, UREA AND OTHER METABOLIC WASTES DIFFUSE INTO THE DIALYSATE. THE SOLUTION IS DRAINED. CYCLE IS REPEATED SEVERAL TIMES.

INTRAOSSEOUS/ INTRAMEDULLARY

- IV ISN'T POSSIBLE
- NEONATES, AVIANS, SMALL PATIENTS WITH CARDIOVASCULAR COLLAPSE, ETC.
- RAPID ABSORPTION THROUGH BONE MARROW
- LONG BONES
- HARD TO MAINTAIN



INTRAVENOUS- IV

- MOST RAPID FOR REHYDRATION
- IV BOLUSES ARE PREFERRED IN SMALL MAMMAL EMERGENCY & AVIAN
- IV CATHETERS ROUTINELY PLACED
  - IN CATS & DOGS FOR EMERGENCY & MAINTENANCE FLUIDS
  - METHOD OF CHOICE FOR ALL FLUID THERAPY IN LARGE ANIMALS.

WHAT CAN WE ADD TO THE FLUID BAG?

COMMON ADDITIVES:

- KCL
- ANTIBIOTICS
- PAIN MEDICATIONS
- ANTIEMETICS
- SEDATIVES
- VITAMINS
- DEXTROSE
- SODIUM BICARBONATE

ADDITIVES

- ALWAYS LABEL BAGS WITH:
  - DATE OPENED
  - TYPES OF ADDITIVES
  - AMOUNT OF ADDITIVES
  - YOUR INITIALS
- FLUIDS THE ARE NOT CONTAMINATED OR DO NOT CONTAIN ADDITIVES CAN BE USED AS SQ FLUIDS.
  - CLINICS WILL VARY AS TO WHAT IS AN ACCEPTABLE AMOUNT OF TIME TO KEEP

STORAGE OF FLUIDS

- CRYSTALLOID AND SYNTHETIC COLLOIDS SHOULD BE STORE AT ROOM TEMP, AND AWAY FROM SUNLIGHT, **WARM BEFORE USE**
- FIRST IN = FIRST OUT. CHECK EXPIRATION DATES
- DISCARD FLUIDS THAT ARE NO LONGER STERILE
  - OPENED BUT NOT USED
  - CONTAINS ADDITIVES
  - ELECTROLYTES/ ADDITIVES PRECIPITATE OR BECOME CLOUDY

- **MAINTENANCE PHASE**
- **REPLACEMENT PHASE**
- **RESUSCITATION PHASE**

*Phases of fluid therapy
*Fluid Calculations

*Maintenance phase

- PROVIDE FLUID SUPPORT TO MAINTAIN CURRENT CONDITION
  - AFTER DEHYDRATION IS CORRECTED, AND THERE ARE NO MORE ONGOING LOSSES.
  - ISOTONIC CRYSTALLOIDS ARE OFTEN USED FOR MAINTENANCE FLUID THERAPY.
  - PATIENTS WITH NORMAL CARDIAC AND RENAL FUNCTION CAN EASILY PROCESS THE EXCESS SODIUM AND ELECTROLYTES IN THE ISOTONIC FLUIDS.
- ANIMALS NEED 50-60MLS H_2O /KG BW/24HRS OR 25-30MLS/#/DAY
  - **60ML/KG/DAY**

*Replacement phase

- CORRECTION OF DEHYDRATION, REPLACEMENT OF ONGOING LOSSES, AND PROVISION OF MAINTENANCE FLUID REQUIREMENTS.
  - ISOTONIC CRYSTALLOIDS MOST COMMON
    - ISOTONIC IN NATURE (VOMITING, DIARRHEA, BLOOD LOSS.)
- ***REPLACEMENT RATE = DEHYDRATION + ONGOING LOSSES + MAINTENANCE***
  - RATE DEPENDS ON SITUATION
    - TOO QUICKLY= NOT ALL ABSORBED
  - RATE IS DETERMINED AFTER THE TIME FRAME FOR REHYDRATION IS DETERMINED.
  - THE TOTAL VOLUME IS DIVIDED BY THE NUMBER OF HOURS DECIDED ON, AND THAT GIVES YOU THE MLS/HR.

RESUSCITATION PHASE (SHOCK)

- RESTORATION OF VASCULAR VOLUME
  - REVERSE HYPOVOLEMIA AND/OR A SHOCK STATE.
- VOLUME ADMINISTERED DEPENDS ON THE VASCULAR VOLUME LOST
  - EFFICIENCY WITH WHICH FLUID EXPANDS IN THE INTRAVASCULAR SPACE.
- HYPOVOLEMIC SHOCK = ~30% LOSS OF BLOOD VOLUME.
  - 25-30ML/KG IN DOGS AND HORSES, AND 12-18ML/KG IN CATS.
  - 3-4ML OF AN ISOTONIC CRYSTALLOID FOR EACH DESIRED 1-ML INCREASE IN VASCULAR VOLUME.
  - SYNTHETIC COLLOID SOLUTIONS: 1 TO 1.5ML FOR EVERY 1ML OF SOLUTION ADMINISTERED.
  - USING MORE THAN ONE TYPE OF FLUID IN A PATIENT AT THE SAME TIME, CAN HAVE SEVERAL POTENTIAL BENEFITS.

*Resuscitation phase

- SHOCK VOLUME DOSES ARE A GUIDELINE
  - TITRATED TO EFFECT BASED ON HOW THE PATIENT IS RESPONDING TO THEM.
  - THIS CAN BE DONE BY GIVING A BOLUS OF FLUIDS, I.E. GIVING ¼ OF THE SHOCK DOSE OVER 10 TO 15 MINUTES.
- MONITOR YOUR PATIENT!!
  - IMPROVEMENT IN HEART RATE
  - CRT (CAPILLARY REFILL TIME)
  - MUCOUS MEMBRANE COLOR (MM'S)
  - BLOOD PRESSURE
  - WARMTH OF DISTAL EXTREMITIES
  - WEIGHT

FLUID REQUIREMENTS

Daily Maintenance

60 X BW (kg)/day

Hydration deficit (ml)

% dehydration × body wt (kg)

% dehydration X BW (lb) X 500ml/ lb

% dehydration X BW (kg) X 1000ml/ kg

On going loss

Estimate total of loss

Replacement rate = dehydration + ongoing losses + maintenance

FLUID REQUIREMENTS

Shock

Crystalloids

- Use for 1-2 hrs or until BP returns to normal
  - Cat 40-45ml / lb /hr
    - 80-90ml / kg / hr
  - Dog 20-30ml / lb / hr
    - 50-70ml / kg / hr

Colloids

- Total dose, administered to effect
- Cat or dog 5-10ml / lb
  - 10-20ml / kg

- BENJI A 25# DOG IS 9% DEHYDRATED AND VOMITED A TOTAL OF 3 CUPS TODAY. CALCULATE HER FLUID RATE FOR THE FIRST 24 HOURS

- MAIN 25# ÷ 2.2= 11.36KG
  - 11.36KG * 55ML/KG/DAY= 625ML/DAY

- ONGOING 1CUP=240ML
  - 240ML*3= 720ML/DAY

MATH OH YEAH!!!!

- REHYDRATE 11.36KG * 0.09 *1000ML= 1,022ML/DAY

- TOTAL REPLACEMENT
- 625ML/DAY+720ML/DAY+1,022ML/DAY= 2,367ML/DAY TO BE REPLACED OVER 24 HOURS 2,367ML/DAY ÷ 24= 99ML/HR

FLUID CALCULATIONS

- CALCULATE **HOURLY FLUID RATE** FOR A 50KG PATIENT THAT IS 9% DEHYDRATED AND HAS VOMITED A TOTAL OF 125MLS.

- CALCULATE **HOURLY FLUID RATE** FOR A 5KG PATIENT THAT IS 6% DEHYDRATED AND HAS VOMITED A TOTAL OF 25MLS.

FLUID OVERLOAD

Heart failure, kidney failure, **technician error**

PULMONARY EDEMA

- LEFT SIDE HEART FAILURE

TACHYPNEA, DYSPNEA

- RESPIRATORY RATE AND PATTERN MONITORED
- INCREASE OF MORE THAN 20%

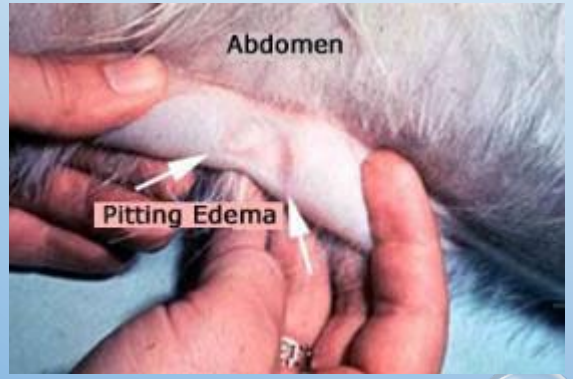
NASAL DISCHARGE

COUGH

CRACKLES

INCREASE IN BODY

WEIGHT (>12-15%)



Cavity effusion or peripheral edema

- Pitting edema

Vomiting

Salivation

Restlessness

Chemosis

Hypertension (>140-150mmHg)

Increased CVP (>8-10 cm H2O)

Urine output

- Normal 1-2ml/kg/hr (not on fluids)
- 2-4ml/kg/hr (on fluids)

END